Ordinary Differential Equation

- **Ordinary Differential Equations (ODE)** relate a function $y(t)$ to its derivatives.
- An **Autonomous** ODE has no explicit dependence on t.
- The order of the highest derivative in an ODE is the **Order**.
- A first order **Initial Value Problem (IVP)** is an ODE with an **Initial Condition (IC)** $y(t_0) = y_0$.
- A function satisfying the IC and the ODE for $a < t_0 < b$ solves the IVP on (a, b).

Example: Solution of a First-Order IVP

Solve the IVP $y' = 2y$, $y(0) = 3$.

The solution of the IVP is $\boxed{y(t) = 3e^{2t}}$

Check: $y'(t) = 6e^{2t} = 2y(t)$ and $y(0) = 3$.

Second Order to First Order

1. Start with a second-order IVP

$$y'' = g(t, y, y'), \qquad y(t_0) = a_1, \qquad y'(t_0) = a_2$$

2. Define $y_1 = y$ and $y_2 = y'$. Then

$$\begin{aligned} y_1' &= y_2 & y_1(t_0) &= a_1 \\ y_2' &= g(t, y_1, y_2) & y_2(t_0) &= a_2 \end{aligned}$$

3. With $\boldsymbol{y} = (y_1, y_2)$ and $\boldsymbol{y}_0 = (a_1, a_2)$, write

$$\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y}), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0, \qquad \boldsymbol{f}(t, \boldsymbol{y}) = \begin{pmatrix} y_2 \\ g(t, y_1, y_2) \end{pmatrix}$$

Mass and Spring

Example: Second Order to First Order

- A mass $m > 0$, damping coefficient $c \geq 0$, spring stiffness $k > 0$, and applied force $g(t)$ give

$$my'' + cy' + ky = g(t)$$

- Displacement $y_1 = y$ and velocity $y_2 = y'$ give

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}'}_{\boldsymbol{y}'} = \underbrace{\begin{pmatrix} y_2 \\ \big(g(t) - cy_2 - ky_1\big)/m \end{pmatrix}}_{\boldsymbol{f}(t,\boldsymbol{y})} \iff \boxed{\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y})}$$

- Initial displacement and velocity supply the two initial values.

Existence and Uniqueness

First-order IVP

If $\boldsymbol{f}$ and its Jacobian $D_{\boldsymbol{y}}\boldsymbol{f}$ (with entries $\partial f_i/\partial y_j$) are continuous near $(t_0, \boldsymbol{y}_0)$, then

$$\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y}), \qquad \boldsymbol{y}(t_0) = \boldsymbol{y}_0$$

has a unique local solution.

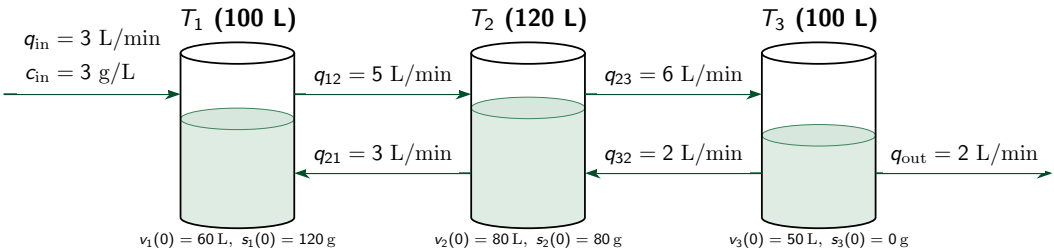
nth-order IVP

If g and its partials wrt $y, y', \ldots, y^{(n-1)}$ are continuous near the IC , then the IVP

$$y^{(n)} = g(t, y, y', \ldots, y^{(n-1)}), \qquad y^{(j)}(t_0) = a_j \quad (j = 0, \ldots, n-1)$$

has a unique local solution.

Well-Mixed Tanks



Tank Volumes

- Volume change is inflow minus outflow, in $\mathrm{L/min}$.

$$v_1' = 3 + 3 - 5 = 1 \qquad v_1(t) = 60 + t$$
$$v_2' = 5 + 2 - 3 - 6 = -2 \qquad v_2(t) = 80 - 2t$$
$$v_3' = 6 - 2 - 2 = 2 \qquad v_3(t) = 50 + 2t$$

- Volumes are in liters. The flow rates do not balance.
- Well-mixed means $c_i(t) = s_i(t)/v_i(t)$, where s_i is salt mass in grams.

When Does the Tank Model Make Sense?

- Each volume must stay positive and below capacity.

$$\begin{aligned} 0 < 60 + t < 100 &\implies 0 \le t < 40 \\ 0 < 80 - 2t < 120 &\implies 0 \le t < 40 \\ 0 < 50 + 2t < 100 &\implies 0 \le t < 25 \end{aligned}$$

- Tank 3 fills first, at $t = 25$ minutes.
- The model applies for $0 \le t < 25$. Overflow changes things!

Salt Flow Rate

- A stream carries salt at rate qc.

$$\frac{\not{\text{L}}}{\text{min}}\frac{\text{g}}{\not{\text{L}}} = \frac{\text{g}}{\text{min}}$$

  - For instance, the external flow brings salt into tank 1 at

$$q_{\text{in}}c_{\text{in}} = \left(3\,\frac{\not{\text{L}}}{\text{min}}\right)\left(3\,\frac{\text{g}}{\not{\text{L}}}\right) = 9\,\frac{\text{g}}{\text{min}}.$$

  - For instance, the pipe from tank 1 to tank 2 brings salt into tank 2 at the rate

$$q_{12}c_1(t) = \frac{5s_1(t)}{v_1(t)} = \frac{5s_1(t)}{\underbrace{60+t}_{\text{volume } v_1(t)}} \quad \text{g/min}.$$

    The same pipe sucks salt out of tank 1 at $-q_{12}c_1(t) = -5s_1(t)/(60+t)\,\text{g/min}$.

Salt Balances

- Salt mass change is salt inflow minus salt outflow.

$$s_1' = 9 + 3c_2 - 5c_1$$
$$s_2' = 5c_1 + 2c_3 - 9c_2$$
$$s_3' = 6c_2 - 4c_3$$

- Substitute the source tank's concentration:

$$c_1 = s_1/(60 + t), \qquad c_2 = s_2/(80 - 2t), \qquad c_3 = s_3/(50 + 2t).$$

- Salt rates are in $\mathrm{g/min}$. Concentrations are in $\mathrm{g/L}$.

Initial Salt Masses

- The diagram gives initial salt masses directly:

$$s_1(0) = 120 \text{ g}, \qquad s_2(0) = 80 \text{ g}, \qquad s_3(0) = 0 \text{ g}.$$

- Concentration is salt mass divided by volume. For example,

$$c_1(0) = \frac{120\,\text{g}}{60\,\text{L}} = 2\,\text{g/L}.$$

- Thus $\boldsymbol{s}_0 = (120, 80, 0)$ in grams.

Tank System as a Vector IVP

$$\boldsymbol{s}' = \boldsymbol{A}(t)\boldsymbol{s} + \boldsymbol{b}, \qquad \boldsymbol{s}(0) = \begin{pmatrix} 120 \\ 80 \\ 0 \end{pmatrix}$$

$$\boldsymbol{A}(t) = \begin{pmatrix} -5/(60+t) & 3/(80-2t) & 0 \\ 5/(60+t) & -9/(80-2t) & 2/(50+2t) \\ 0 & 6/(80-2t) & -4/(50+2t) \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 9 \\ 0 \\ 0 \end{pmatrix}.$$

- $\boldsymbol{A}(t)$ varies with time because the volumes change.
- Entries of $\boldsymbol{A}$ have units min^{-1}. Entries of $\boldsymbol{b}$ have units $\mathrm{g/min}$.
- The physical model applies for $0 \le t < 25$ minutes.

Total Salt Balance

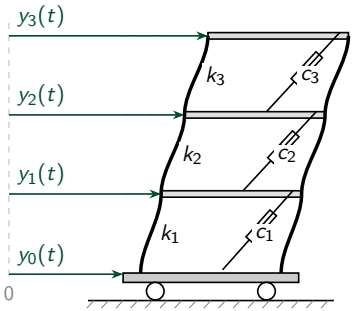
- Add the three salt balances. Internal transfers cancel.

$$(s_1 + s_2 + s_3)' = 9 - 2c_3 = 9 - 2s_3/(50 + 2t).$$

- Total salt changes only through the external inlet and outlet.

Three-Story Building

Displacements: $\boldsymbol{y}(t) = (y_1(t), y_2(t), y_3(t))$ (m), t in seconds.



$m_3 = 2\,\mathrm{kg}$
$k_3 = 2\,\mathrm{N/m}$
$c_3 = 1\,\mathrm{N\,s/m}$

$m_2 = 3\,\mathrm{kg}$
$k_2 = 3\,\mathrm{N/m}$
$c_2 = 1\,\mathrm{N\,s/m}$

$m_1 = 4\,\mathrm{kg}$
$k_1 = 4\,\mathrm{N/m}$
$c_1 = 2\,\mathrm{N\,s/m}$

$y_0(t) = \sin(t)\ \mathrm{m}$

Elastic and Damping Forces Between Floors

1. Displacements $y_i(t)$ in m rightwards from origin.
2. Between floors $i-1$ and i:

$$\underbrace{y_i - y_{i-1}}_{\text{relative displacement } [\mathrm{m}]} \qquad \underbrace{y_i' - y_{i-1}'}_{\text{relative velocity } [\mathrm{m/s}]}$$

3. Walls transmit forces

$$f_i = \underbrace{k_i(y_i - y_{i-1})}_{\text{elastic}} + \underbrace{c_i(y_i' - y_{i-1}')}_{\text{damping}} \quad [\mathrm{N}]$$

4. Walls exert forces $-f_i$ on floor i and $+f_i$ on floor $i-1$.
5. Level 0 is the driven base. Its walls exert force on floor 1

$$-f_1 = 4(y_0 - y_1) + 2(y_0' - y_1') \quad [\mathrm{N}].$$

Force Units Newton says $f = ma$

1. Spring force in N is stiffness (k_i $\mathrm{N/m}$) times relative displacement ($y_i - y_{i-1}$ m)

$$k_i(y_i - y_{i-1}): \qquad \frac{\mathrm{N}}{\cancel{\mathrm{m}}}\cancel{\mathrm{m}} = \mathrm{N}$$

2. Damping force in N is damping (c_i $\mathrm{N\,s/m}$) times relative velocity ($y_i' - y_{i-1}'$ $\mathrm{m/s}$)

$$c_i(y_i' - y_{i-1}'): \qquad \frac{\mathrm{N}\,\cancel{\mathrm{s}}}{\cancel{\mathrm{m}}}\frac{\cancel{\mathrm{m}}}{\cancel{\mathrm{s}}} = \mathrm{N}$$

3. Force in N is mass (m_i kg) times acceleration (y_i'' $\mathrm{m/s^2}$)

$$m_i y_i'': \qquad \mathrm{kg}\,\frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

Floor Balances

1. Net force in N is mass times acceleration

$$4y_1'' = -f_1 + f_2, \qquad 3y_2'' = -f_2 + f_3, \qquad 2y_3'' = -f_3$$

2. Use the displayed stiffness and damping values

$$\begin{aligned} f_1 &= 4(y_1 - y_0) + 2(y_1' - y_0') &&= 4y_1 - 4y_0 + 2y_1' - 2y_0' \\ f_2 &= 3(y_2 - y_1) + (y_2' - y_1') &&= 3y_2 - 3y_1 + y_2' - y_1' \\ f_3 &= 2(y_3 - y_2) + (y_3' - y_2') &&= 2y_3 - 2y_2 + y_3' - y_2' \end{aligned}$$

3. Collect floor terms on the left and base terms on the right

$$\begin{aligned} 4y_1'' + 3y_1' - y_2' + 7y_1 - 3y_2 &= 2y_0' + 4y_0 \\ 3y_2'' - y_1' + 2y_2' - y_3' - 3y_1 + 5y_2 - 2y_3 &= 0 \\ 2y_3'' - y_2' + y_3' - 2y_2 + 2y_3 &= 0 \end{aligned}$$

Floor Equations Continue

$$\begin{aligned} 4y_1'' + 3y_1' - y_2' + 7y_1 - 3y_2 &= 2y_0' + 4y_0 \\ 3y_2'' - y_1' + 2y_2' - y_3' - 3y_1 + 5y_2 - 2y_3 &= 0 \\ 2y_3'' - y_2' + y_3' - 2y_2 + 2y_3 &= 0 \end{aligned}$$

$$\implies \begin{array}{llll} 4y_1'' & + 3y_1' - y_2' & + 7y_1 - 3y_2 & = 2y_0' + 4y_0 \\ 3y_2'' & - y_1' + 2y_2' - y_3' & - 3y_1 + 5y_2 - 2y_3 = & 0 \\ 2y_3'' & - y_2' + y_3' & - 2y_2 + 2y_3 = & 0 \end{array}$$

$$\implies \begin{array}{llll} 4y_1'' + 0y_2'' + 0y_3'' & + 3y_1' - y_2' + 0y_3' & + 7y_1 - 3y_2 + 0y_3 = 2y_0' + 4y_0 \\ 0y_1'' + 3y_2'' + 0y_3'' & - y_1' + 2y_2' - y_3' & - 3y_1 + 5y_2 - 2y_3 = & 0 \\ 0y_1'' + 0y_2'' + 2y_3'' & + 0y_1' - y_2' + y_3' & + 0y_1 - 2y_2 + 2y_3 = & 0 \end{array}$$

Second-Order Matrix System

- Force balance is

$$\boxed{\boldsymbol{M}\boldsymbol{y}'' + \boldsymbol{C}\boldsymbol{y}' + \boldsymbol{K}\boldsymbol{y} = \boldsymbol{b}(t)}$$

where

- $\boldsymbol{M}$, $\boldsymbol{C}$, and $\boldsymbol{K}$ have units kg, $\mathrm{N\,s/m}$, and $\mathrm{N/m}$ respectively

$$\underbrace{\boldsymbol{M} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}}_{\text{mass matrix}} \quad \underbrace{\boldsymbol{C} = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}}_{\text{damping matrix}} \quad \underbrace{\boldsymbol{K} = \begin{pmatrix} 7 & -3 & 0 \\ -3 & 5 & -2 \\ 0 & -2 & 2 \end{pmatrix}}_{\text{stiffness matrix}}$$

- The external force vector $\boldsymbol{b}(t)$ has units N

$$\boldsymbol{b}(t) = \begin{pmatrix} 2y_0'(t) + 4y_0(t) \\ 0 \\ 0 \end{pmatrix}$$

First-Order System (Six Scalar Equations)

1. Define velocity $\boldsymbol{v} = \boldsymbol{y}'$ in $\mathrm{m/s}$. Stack displacement and velocity into the six vector

$$\boldsymbol{z} = \begin{pmatrix} \boldsymbol{y} \\ \boldsymbol{v} \end{pmatrix} \qquad \boldsymbol{z}' = \begin{pmatrix} \boldsymbol{y}' \\ \boldsymbol{v}' \end{pmatrix}$$

2. Isolate acceleration, then multiply by $\boldsymbol{M}^{-1}$

$$\boldsymbol{M}\boldsymbol{y}'' = -\boldsymbol{K}\boldsymbol{y} - \boldsymbol{C}\boldsymbol{y}' + \boldsymbol{b}(t)$$

$$\boldsymbol{y}' = \boldsymbol{v} \qquad \boldsymbol{v}' = \boldsymbol{y}'' = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y} - \boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v} + \boldsymbol{M}^{-1}\boldsymbol{b}(t)$$

3. The top three equations give velocity. The bottom three give acceleration

$$\boldsymbol{z}' = \begin{pmatrix} \boldsymbol{y}' \\ \boldsymbol{v}' \end{pmatrix} = \underbrace{\begin{pmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{M}^{-1}\boldsymbol{K} & -\boldsymbol{M}^{-1}\boldsymbol{C} \end{pmatrix}}_{\boldsymbol{A}} \underbrace{\begin{pmatrix} \boldsymbol{y} \\ \boldsymbol{v} \end{pmatrix}}_{\boldsymbol{z}} + \begin{pmatrix} \boldsymbol{0} \\ \boldsymbol{M}^{-1}\boldsymbol{b}(t) \end{pmatrix}$$

Here $\boldsymbol{0}$ and $\boldsymbol{I}$ are the 3×3 zero and identity matrices.

Acceleration Units

- Acceleration in $\mathrm{m/s^2}$ is force (N) divided by mass (kg)

$$\frac{\mathrm{N}}{\mathrm{kg}} = \frac{\cancel{\mathrm{kg}}\,\mathrm{m}}{\mathrm{s}^2\,\cancel{\mathrm{kg}}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

- Every term in $\boldsymbol{v}' = -\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y} - \boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v} + \boldsymbol{M}^{-1}\boldsymbol{b}$ has these units

$$\underbrace{\frac{1}{\cancel{\mathrm{kg}}}\frac{\cancel{\mathrm{kg}}}{\mathrm{s}^2}\,\mathrm{m}}_{\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{y}} = \frac{\mathrm{m}}{\mathrm{s}^2}, \qquad \underbrace{\frac{1}{\cancel{\mathrm{kg}}}\frac{\cancel{\mathrm{kg}}}{\mathrm{s}}\frac{\mathrm{m}}{\mathrm{s}}}_{\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{v}} = \frac{\mathrm{m}}{\mathrm{s}^2}$$

First-Order Building IVP

$$\boldsymbol{z}' = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{r}(t) = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -7/4 & 3/4 & 0 & -3/4 & 1/4 & 0 \\ 1 & -5/3 & 2/3 & 1/3 & -2/3 & 1/3 \\ 0 & 1 & -1 & 0 & 1/2 & -1/2 \end{pmatrix} \boldsymbol{z} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ y_0 + y_0'/2 \\ 0 \\ 0 \end{pmatrix}$$

- Initial positions ($\boldsymbol{y}(0)$ m) and velocities ($\boldsymbol{v}(0)$ m/s) give

$$\boldsymbol{z}(0) = \begin{pmatrix} \boldsymbol{y}(0) \\ \boldsymbol{v}(0) \end{pmatrix}$$

- Prescribing base motion $y_0(t)$ and the six values for $\boldsymbol{z}(0)$ completes the IVP.